

NIZHNEUSINSKOYE, Russian Federation

WMO: 360910

Lat: 52.257N

Lon: 93.013E

Elev: 662

StdP: 93.62

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.5	(c) -36.8	(d) -42.3	(e) 0.1	(f) -39.2	(g) -39.5	(h) 0.1	(i) -36.6	(j) 4.9	(k) -24.9	(l) 3.0	(m) -25.4	(n) 0.2	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	30.2	17.7	28.0	17.4	26.0	16.5	19.4	26.5	18.4	25.2	17.5	23.9	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.4	21.6	16.2	12.5	20.5	15.2	11.7	19.7	58.5	26.5	55.1	25.3	52.2	24.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.0	(b) 5.3	(c) 4.1	(e) -41.8	(f) 33.9	(g) 2.0	(h) 2.2	(i) -43.2	(j) 35.5	(k) -44.4	(l) 36.8	(m) -45.5	(n) 38.1	(o) -47.0	(p) 39.7		
(4) 7.0	(5) 5.3	(6) 4.1	(7) -42.0	(8) 21.2	(9) 1.9	(10) 1.7	(11) -43.3	(12) 22.4	(13) -44.4	(14) 23.4	(15) -45.5	(16) 24.4	(17) -46.9	(18) 25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-26.1	-19.7	-6.3	3.9	8.8	15.5	17.9	15.6	8.4	-0.3	-13.6	-22.5		
	DBStd	15.87	4.90	6.01	6.19	4.27	3.91	4.02	2.67	3.04	4.04	4.76	6.68	6.13		
	HDD10.0	4828	1118	831	506	187	69	5	0	1	77	318	707	1008		
	HDD18.3	7266	1377	1064	764	434	296	101	40	93	297	577	957	1267		
	CDD10.0	654	0	0	0	3	32	171	244	175	30	0	0	0		
	CDD18.3	50	0	0	0	0	0	16	26	8	0	0	0	0		
	CDH23.3	1147	0	0	0	2	67	439	432	188	18	0	0	0		
	CDH26.7	348	0	0	0	0	15	156	138	38	2	0	0	0		
	WSAvg	0.8	0.2	0.4	0.8	1.6	1.8	1.2	1.1	0.8	0.8	0.5	0.3	0.2		
	PrecAvg	723	25	25	39	48	61	78	105	101	68	68	61	50		
	PrecMax	956	62	82	108	102	142	130	245	203	110	306	147	156		
(15) Precipitation	PrecMin	489	2	1	5	13	1	16	5	56	35	16	17	3		
	PrecStd	113	15	19	26	27	35	31	38	34	20	55	33	37		
	DB	-12.4	-1.8	12.4	22.9	28.9	33.8	33.2	30.0	25.6	15.9	2.5	-6.5			
	MCWB	-13.1	-3.7	5.1	10.9	14.8	17.6	19.8	19.2	15.7	9.1	0.4	-7.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-14.9	-5.3	8.8	18.8	25.1	30.5	30.1	27.6	22.4	12.2	0.2	-9.0			
	2% MCWB	-15.5	-6.9	3.2	9.2	13.2	17.1	18.1	18.3	14.2	7.0	-1.2	-9.5			
	5% DB	-16.8	-7.8	6.2	16.0	22.1	28.2	27.6	25.3	19.6	9.2	-2.2	-11.4			
	5% MCWB	-17.2	-9.1	1.5	7.8	11.7	16.6	17.8	17.1	12.9	5.2	-3.6	-11.9			
	10% DB	-18.5	-10.4	3.4	13.4	18.9	25.6	25.5	23.2	16.8	6.6	-4.5	-14.1			
	10% MCWB	-18.9	-11.4	-0.3	6.4	10.5	15.8	17.1	16.3	11.5	3.8	-5.6	-14.5			
	0.4% WB	-12.9	-3.5	5.2	11.5	15.6	19.7	21.0	20.7	16.5	9.9	0.9	-7.3			
	0.4% MCDB	-12.3	-2.1	11.5	21.1	27.5	27.6	29.7	26.9	23.7	14.4	2.2	-6.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-15.4	-6.9	3.5	9.8	14.0	18.4	19.7	19.2	14.8	7.3	-1.3	-9.5			
	2% MCDB	-15.0	-5.4	8.4	18.1	23.6	27.1	26.8	25.7	21.2	11.0	-0.1	-9.0			
	5% WB	-17.1	-9.1	1.7	8.0	12.5	17.4	18.7	17.9	13.4	5.6	-3.3	-11.9			
	5% MCDB	-16.8	-8.0	5.9	14.9	20.7	26.0	25.1	24.1	18.7	8.9	-2.2	-11.4			
	10% WB	-18.9	-11.4	-0.1	6.7	10.9	16.5	17.9	16.7	11.8	3.9	-5.5	-14.4			
	10% MCDB	-18.5	-10.5	3.2	13.1	18.0	24.2	23.6	21.9	16.2	6.3	-4.4	-14.1			
	MDBR	11.1	14.7	16.6	15.2	16.1	16.4	14.1	13.9	15.4	11.4	9.7	8.9			
	5% DB	10.0	14.4	16.9	20.5	23.4	22.1	19.5	18.9	20.5	16.6	9.5	8.0			
(36) Mean Daily Temperature Range	5% DB	9.4	12.5	11.2	11.6	12.3	9.6	9.0	10.1	12.1	10.8	7.6	7.6			
	5% WB	9.7	14.1	16.2	19.2	21.3	18.4	15.6	16.9	18.9	15.9	8.9	7.8			
	MCWBR	9.3	12.3	10.8	11.1	11.4	8.2	8.0	9.4	11.3	10.4	7.3	7.4			
	MCWBR	9.3	12.3	10.8	11.1	11.4	8.2	8.0	9.4	11.3	10.4	7.3	7.4			
(40) Clear-Sky Solar Irradiance	taub	0.276	0.311	0.361	0.405	0.413	0.407	0.404	0.388	0.352	0.339	0.311	0.281			
	taud	2.068	2.021	1.957	2.092	2.169	2.302	2.348	2.362	2.396	2.284	2.121	2.026			
	Ebn at Noon	691	774	805	823	837	847	843	837	828	748	637	596			
	Edh at Noon	76	109	144	143	139	123	116	109	93	84	73	67			
(44) All-Sky Solar Radiation	RadAvg	1.24	2.32	3.84	4.84	5.48	6.03	5.47	4.72	3.53	2.12	1.14	0.84			
	RadStd	0.08	0.12	0.17	0.23	0.39	0.50	0.35	0.36	0.35	0.19	0.12	0.07			

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page